

GAP: A study on core reconstruction bias and how it attenuates the early-late asymmetry

Lautaro Silva Pizzi, Juan Manuel Figueira and Federico Sánchez
Pierre Auger Observatory

August, 2026

Abstract

- 1 Introduction**
- 2 Phenomenological Context and Methodology**
- 3 Motivation: The Attenuation of the Reconstructed A_1 Amplitude**
- 4 Evaluation of Reconstruction Biases: Directional Counting vs. Geometric Core Shift**
 - 4.1 The Directional Counting Bias and the Empirical Toy Model**
 - 4.2 The Geometric Core Shift: The Symmetric LDF Conflict**
 - 4.3 Coupled Degradation Effects**
- 5 Summary and Conclusions**

References